

ActInf Textbook Group ~ Cohort 9 ~ Session 6

(Chapter 4) ~ 9/25/2025

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All right, welcome back everyone. It's September 25th and we're in chapter 4. So, does anyone want to start with any part of chapter 4? any section or quotation or question that that they have or that was already written down by a previous participant or just any part of four to jump into. I have a quick question just about notation and conventions. Um, and this is just to sort of orient me to >> Wait, you you muted there for a second. Would you just orient to me to which domains this particular notation is most associated with >> statistics? >> Okay. >> Y being observation and X being latent state is statistics. And then there are some slight differences between like the mathematical control theory and the engineering applied control theory for like which variable they might use for a transition matrix or for a control variable. But it's it's largely grounded in in this relationship between observables and unobservables. Any other just random interesting quote from the chapter or a question? Or we can look at some of the previous questions or we can skim through the chapter. Okay, let's kind of just go through the chapter, but write in the chat or raise your hand if you if you want to um proceed. Okay, starts with everything should be made as simple as possible but not simpler. But isn't that the question? And the good news is that's where the basian information criterion comes into play which is like rewarding things for fitting but then penalizing them for having more parameters because you always fit better by having more parameters in the model but then there's some kind of bend where you're getting diminishing returns. So that's a theme that comes up again and again is like how do you trade off those dual imperatives of fitting well but not overfitting or simplifying but not oversimplifying. Okay, introduction paragraph complements the preceding chapter's conceptual treatment with a more formal treatment which is kind of a funny way to begin the chapter because it's like wait after chapter two and three a more formal treatment. It's like yes. Um but now in terms of the low road and the high road which were covered in chapters two and three, now we're in chapter 4. We took both of the approaches to the place. Now this is where the actual generative models are going to be

described. A key focus is on different ways of handling time, specifically time as discrete time steps or continuous flow which gets revisited in chapter seven and eight. Okay. 4.2 picks up with the notion of the Bayesian brain which is uh one of the sort of family slash predecessor theories of just thinking about what organisms or brains do in terms of Bayesian inference problems which get broken down into these two broad categories of cognition, inbound perception doing something about inferring the state of the world and then action specifically um because exact inference even on a computer is often infeasible. There's all kinds of heuristics and even if the computation were feasible on a computer, we would it it's basically just a foregone conclusion to say that's well like quote that's not what the organisms really doing. It's like right it's a computer program. It's not what the organism is doing. Um, so they're going to highlight why the use of free energy statistics are useful in these generative models and what those models are about. The chapter is more technical than 1.3 and some of these topics are in the appendices, others are just a good side readings. But linear algebra, so systems of equations with variables that are fixed or or known or unknown, differentiation, looking at integrals and derivatives, and the Taylor series expansion for the continuous time setting of moving further and further away, estimating a function through using derivatives. they return to B theorem and just write it out. X is a latent state. Y is an observation. And there's all different kinds of ways to kind of reconfigure B equation depending on which parts you know and which parts you don't know. But that was visited in chapter 2 with the frog and the apple. Okay. uh here takes an interesting turn into some of the fundamental properties of logs. So I think we will have a yeah so it's often brought up like why do why use a probability versus a log probability? So there's a few computer science reasons and then there's quite a number of analytical reasons. So this is a good link but this this kind of information can be found anywhere which is these these kinds of calculations make log probabilities very amenable to statistics. So the log of a product is the sum of the logs. So if we wanted to have the probability of a coin flip and rolling a die those are independent. So we want to multiply the probabilities but then the logs get summed. So there's calculations that are independent can just be summed in logs and and um similarly for divisions and subtraction. Um exponentiation is well behaved in logs. And then there's some computer science considerations like if it's a very small number you want it's easier to store a large number than a tiny number on a computer like 1 billion rather than 1 billion. And so if we're dealing with low with given events that have low probability it can be easier to store that as a log probability. Um, and then there's this Jensen's inequality which is the log of an average is greater or equal to the average

of a log. And that is going to come into play through these steps described here. um basically as as a confirmation that we'll be able to uh approach the log probability which is the value we actually care about and have that as a bound for a free energy value. That's an interesting section. We can come back to it but returning carry on. So basically there's there's a divergence that can be calculated separately from the overall level of likelihood. that matters because in terms of incremental sequential opt optimization what's really important is that there's a procedure to improve step by step not necessarily that you have the best estimate of the overall exact likelihood of something happening. So that's primarily looking for attractable evidence maximization scheme. So that's the KL divergence and and there's some other choices there, but that's um just kind of revisiting that foundational idea. Okay, this is a really this is kind of the key section of the key chapter and it relates to a lot of the questions that people have about what is free energy and a lot of times there's there's ambiguity of like well what what about this situation and what is the free energy of this fly in that situation and it's like this is an example of a quote that makes it very clear that free energy is a property of models not of systems. So free energy is a characteristic of maps not territories. That's kind of this core um meme that helps forestall a lot of other nonsequators. To calculate the free energy we need three things data a family of distributions and a generative model which is the prior and likelihood. So free energy is a calculated statistic with those things in hand. If those values are in hand, you can make this calculation. If the values are not in hand, it would be like saying, well, what's the regression co coefficient or what's the sum of squares of this linear regression? But I'm not going to tell you what the data are. So, it's kind of like an incomplete sentence, but it's it's it's just interesting that this is the narrow form of of um what free energy is, but that that helps contextualize like how broadly people kind of take it beyond that. All right. They are going to highlight a few different generative models in this section. There are models with categorical variables in their state spaces and there are models with continuous variables in their state spaces. And then there's this other axis which is whether time is handled as a discrete sequence of events or whether time is handled as a continuous differentiable flow. So kind of explored that before, but there's like 2 by two quadrants in terms of this taxonomy of generative models and you can have hybrids that that include all all four quadrants or whatever you want. But just at the first pass, there's discrete time and continuous time. And then within either of those time handlings, you can either have a discrete categorical variable um like hot, medium or cold. Or you could have a continuous variable in the state space like on a scale of 1 to 100. Okay, figure 4.2 is a really nice background figure that that um you

know, again, there's infinite ways to reconfigure these textbooks. The whole point is there there's not a linear path, but this is a great background concept to get a better handle on what these graphs that we're going to see are all about and to connect it to causal reasoning. So this is kind of the top left is like there's a latent variable that gives rise to an observable and then you can talk about well what what is the what is the raw py what's just the distribution over y's like and then there's what is the distribution over y conditional upon x like what's the overall temperature distribution and then there's what's the temperature distribution conditional upon it being daytime or night time um And then there's other causal settings like there could be one latent factor that causes two observations. There could be two causal factors that cause one observation. There could be a chain of causal consequences that lead to an observation. So this is just like thinking compositionally or graphically about causal flows in these kinds of structural models. Um, in the parametric statistic setting, this is often called a structural equation. So they'll take a big data set and then they'll they'll graph out like connections just based upon kind of correlations or like if you account for this correlation then this correlation becomes stronger or weaker. Um so those are like kind of corrected or calibrated correlation statistics and then what causal logic or what graphical modeling does. This is the kind of pearl direction is is just explicitly include causal factors and observations. So again, that's what these graphs are showing is you have Yeah. Go ahead. >> Oh. Um, what is the way to get from an equation or a data set to drawing one of these diagrams? like are they visual sort of aids or are they like actually capturing uh something mathematical and like there's a right and a wrong way. So to determine causal diagrams for data sets is kind of like it's sometimes called structured learning. But that's kind of the the the that would be like the generic superpower would be to take data like numerical data and then propose causal models. Now just like the quote at the beginning of the chapter, there's potentially an infinite number of causal models depending on which variables you want to include or not. So it isn't just that there's one but then on the other hand sometimes you explicitly want to specify the causal model and then you want to generate synthetic data and then with that synthetic data you want to see whether you can recover the parameters like determine that you did have the correct causal model underlying it. But that is this whole area of causal discovery and modeling is basically how do you draw those edges among empirical variables and then how do you propose and connect up other latent variables to make the most robust performance account. And then it's like, well then do you want fewer variables that explain the data less well or do you want more variables that explain it better? And that's where that basian information

criterion comes into play because you're kind of comparing a portfolio of models and then they can be compared in different features like um how sensitive are they to the amount of data or to the variability of the data or how models that have fewer or more parameters. Yeah, Benjamin. Okay, so I mentioned I'm like one or two chapters ahead, but without context in mind like this. I like this uh diagram. It's showing some causality between these variables later and this is the question like later I think in the book it says the generative model itself may have priors about your beliefs of on causal structures like this right so uh it this is like applying it to the base equation which is like pretty clear we we all believe we all we all have the same beliefs about what the causal structure of bay law is but is it also true that like this concept's useful Because when designing models later, the model also might have in its priors beliefs about causal structures that are more tenuous or about less known structures than bas law. Does that make sense? >> Yeah. Well, one of the great parts about basian modeling, whether it's used as the scientific tool to specify models or whether you're talking about making an agent who is a basian causal reasoner and will be proposing its own possible distributions is like it makes the priors explicit. you can see where something has just been fixed like and the buck does have to stop somewhere and you can see where that is like the prior of it being um day or night is just 50/50 and then someone else says oh it's not 50/50 I think it's actually drawn from this other distribution okay well what is your distribution on that so like a hyper hyper prior or a hyper parameter but eventually you're going to get to some distribution that you're drawing the highest hyper prior from. Even if it's a uniform distribution or if it's what whatever shape it is, but it it has to be drawn from somewhere or be a fixed number. And then that's kind of the fun about the delta distribution. It's it's a quirky distribution, but it's basically just a spike on the line. So that's like this the it's it's it's a very unique distribution. It's basically just a number, but it's like a number wrapped up like a statistical distribution which lets you say okay um and my my prior for the number of buses per day is eight period. And then that gets all the semantics of being a full distribution even though it's literally just the number eight. this graph. They keep on discussing it and then they pivot to this kind of core Rosetta Stone image which is also the graphical abstract of what's to come in chapters seven on the top and eight which is like okay so that was the general setting of latent variables and observations and the ways that we can have these kind of um operators that can do all the stuff that we would expect with Legos. does like do a transformation or merge or split. Well, now that's going to get brought into play with these graphical models of a simple cognitive system and uh yes Anna wrote circles versus square are relevant for

understand. Yeah, great great point. So the circles are random variables themselves and then the squares are distributions that have the given um meaning. So like one here is a probability distribution on X giving rise to the random variable X . And this is going to come back again with the message passing and the factor graphs and and so it can be revisited at a deeper level. But yes, like on on the uh top graph here, we have several circles and then several squares. The circles are the random variables themselves and then the squares are kind of um distributions that get specified when you use the code example you specify all of them. So in I mean just in practice it's not like circles and squares. They're all variables in a statistical model but for clarity they're they're being drawn out this way. Um they talk about why they have this different notation between the top and the bottom even though there's this provocative juxtaposition to show that they have a overall congruent shape and the variables are placed and they're doing the same thing roughly um in the top and the bottom model the the backbone of this this is the discrete time model. The backbone of this model with the s is latent states changing through time. So s of s of sub t is latent state in the moment. Then we have the next moment and the previous moments. Three is a distribution that says what is the latent state at the next moment conditional upon the state at this moment and the action being selected. So that's why three is getting inserted there. So that's how you got from $t-1$ to t . It's how you get from $t-1$ to $t+1$. So that's the kind of back and then the initial three is your prior over S . So that just kicks off the chain. But you only need one kind of initiator to the chain and then it's just going to be this markoff transition chain of latent states. That's the kind of core variable like temperature changing through time sequence of temperature variables. But then there's two parts. there's like a above and a below. So at each time point looking below there's going to be matrix 2 leading to an observation. So this is the two is the distribution of observations at that time given latent state at that time. So if it was temperature changing through time in the room these would be like thermometer measurements. So in practice, it looks like you have a sequence of temperature observations on your data sheet, but the way that's being modeled is there's a latent state that's changing through time and at each time it's emitting an observation. So it's not that you have a chain, a markov chain of observations. You have a markov chain of latent states and each of them are going to emit an observation. So that's the downstairs part is the prior and then the time dynamics of a latent state and then how latent states emit observations at each time point and that gets explored like in the music example in chapter 7 and then there's the upstairs part. So this is the sort of sense making and perception part which is the relationship between observations and latent states.

And then the control theoretic part or the action inference part is instead of just having a single matrix applied again and again and again for the transition matrix of between time points on s_{pi} is like the ability to choose between different transition matrices. So instead of just one matrix being applied again and again which is how a simple markoff chain would happen. It's like there's multiple index cards that can get selected and then action selection is which transition matrix is going to get selected to play the role of state transition in the next time step. And then how you determine the probability of different actions is the whole question of action inference. And one way to do that which is the reward uh learning or reinforcement learning approach would be we're going to associate different actions or different action state pairs with a value function and then we will use the value function to either select the best action most valued action or we'll use the value uh value function to reweight the distribution of action probabilities and select from that. So that is how most reinforcement learning approaches work. The difference and what is going to get explored in the expected free energy is actually a more sophisticated way of directly working with the probabilities of which obviates the need to specify an auxiliary value function or reward function. Rather actions are just directly evaluated in terms of their pragmatic value which is their alignment with preferred observations and epistemic value which is their expected information gain. And then how different policies score on expected free energy epistemic plus pragmatic value is used to reweight the probabilities of those actions being taken. And then that direct probability of those actions is used to either select the most likely action or draw from that distribution to select which transition action to take. That's kind of at the core of the difference in action selection between reinforcement learning approaches and active events.

the bottom graph, the nodes are going to play kind of cognitively slash functionally the exact same roles, but it's important to note how the math and the notation are different. So here the backbone part instead of being a sequence of latent variables and their time evolution it's actually just one variable then its derivative and its second derivative and etc. So we have like the temperature in the room through time the rate of change of the temperature in the room through time. the rate of change of the rate of change of the temperature through time. So whereas three was playing the role of a transition matrix in the discrete time setting, three in the continuous time setting is playing the role of basically a differentiation operator. Two, the mapping between latent states and observations is roughly similar, but just noting that the observations are not observations at t minus one, current moment in the future. They are the observations through time, the rate of change of the observations and the second rate of change. And then this top

part is uh again playing the functional role of action selection. But whereas pi was playing this kind of procedural action selection intervention role like turn on the air conditioner or turn off the air conditioner. Here v is controlling the differentiation So whereas like if we're let's say we're in a car and we have forward or backwards or stay the same that would be modeled as like three different actions you could take that would increase your speed or decrease it or keep it the same. that could also be encoded, you can imagine as three ways to in to make a derivative from the speed to the d to the rate of change of the speed. Like if you are putting your foot on the gas, the acceleration is plus one, then that will propagate through the speed increasing. Or if you're putting on the brake, the acceleration is minus one. they're being laid out. It's just a uh graphful abstract for chapters seven and eight. And it's just highlighting that there are these two different ways to handle time, but that graphical models have the flexibility to accommodate both those settings. I have a question. >> Yep. >> So I think I tracked that for the most part except except for intuitively I want to make sure I understand the second graph like these line these horizontal lines are going out further than just three nodes right through in this case two or three derivatives right like that variable would be going through like even more derivatives as the calculation goes on. Is that is that am I right in understanding that like there's only these diagrams are only three wide on the x- axis right now but like it models an equation that would go much w much much further and I start to lose my intuition for what the 50th derivative would be here. >> Yeah. >> And why that's useful like why that's useful. I get the the first one a lot more than the second one. But I I lose my intuition once you're doing that many derivatives. Like what intuitively does that correspond to for the agent? >> Okay. So on the top um of the examples, it corresponds to time steps. So you could have an agent that's just considering one time step ahead. Even if the simulation runs for a thousand, it's just doing two. >> That makes sense to me. >> Or Yeah. or the the agent's explicit horizon could be longer and so okay then now this question about the generalized coordinates on on the bottom. So ultimately you can have the model with just one or two or three or however many coordinates. Um, this was a Let me show you what I >> mean. The first thing you said was kind of helpful. Is Yeah. is the 50th derivative. It's just the continuous time equivalent to thinking 50 ticks ahead in the >> Yeah. Yeah. Yeah. >> Yeah. That is the um Okay. So, I I was playing with this in the um in the RX infer toolkit and basically Okay. I think it's in meta analysis and basically looked at how well does it fit. Okay. Yeah. So, looked at how well Um so there's um the rows are how many orders of generalized coordinates do you have? One so just the just the rate of change just you know first derivative second derivative etc. 1 through eight, the eighth

derivative, you know, then there's three scenarios. There's a constant scenario that was just like a straight line. Then there was a piece-wise mixed scenario with like a bunch of weird stuff happening and then a sign. And so it's like these are the kinds of diagnostic plots that you can make to empirically assess, okay, I'm getting diminishing returns. Like for example in the constant setting they did no better with with more. So in practice you are only having and then like Kristen has been asked this many times like how many orders and then he'll say something like six just like to joke but it but it's empirical how many orders you want to include. Yeah. Based upon how how Yeah. So it's basically a structured learning question. Is it worth the extra compute? Am I getting interpretability? PID control which is the the standard engineering you know for for heating and airplanes and stuff like that is just three. So doesn't but it's cool that it exists within this generalized setting but but it doesn't need to go that many deep. Okay. Le wrote um with a difference in policy between reinforcement learning and active inference can active inference agent do inmodel simulations like in planning? Yes. Expected free energybased policy inference is based upon explicit policy rollouts. I'm not sure if that addresses it but that is how planning as inference is done which is kind of like a chess just in the simplest form. It just considers all the actions at each time step and what would be the consequences of those actions over the time horizon and then evaluates each of those sequences of actions as policy sequences and uses that to calculate expected free energy uses the expected observations of those um sequences. >> Oh, sorry. Then I I I still don't see the difference in policy between environmental learning and active influence in that case because to me it seems like they both sample policy from the distribution if we're in the pending case. Is that right? >> Yeah. So they they are going to have the same policy space. So let's just say we have um left or right and we have a planning horizon of four. So both a reinforcement learning and an active inference approach are going to consider the same set of policies. But the reward learning approach is going to associate a value with each of those. There's going to be another value function. Basically then there's going to be a map between each policy and its score and then this policies will be selected based upon their scoring on this other proposed distribution which could have been trained with data or specified ad hoc. Whereas in the active inference approach, those policies are being directly evaluated for their epistemic and pragmatic value and then those direct evaluations are used to update the likelihood of that policy happening. So still if you're dealing with a with a policy planning agents it will look ahead and it will consider the exact same policy space. But policies are being compared and updated in their likelihood based upon this measure which doesn't invoke

any other distributions like a score or a reward. But the score reward can be a super useful strategy. If you have if you have expert trajectories and and negative examples, you can use supervised training to say, well, these are bad, these are good. So, make a score function that makes more policies that look like this and less that look like that. And then you don't need to explicitly consider the the pragmatic value or epistemic value of policies individually. Does that address it? >> Yeah. I just think that the difference is smaller than I imagined. Like the choice of the prior here in the pragmatic value it seems like it can be matched one to one to the choice of the reward model in transform learning. >> Yes. Yeah. Exactly. Exactly. Yes. But you could do that reinforce reward learning or utility maximization is a special case. If you don't need epistemic value, then you don't necessarily need to engage with any of the epistemics of active inference. If you're just looking for a utility maximizer on a fixed problem, then you're in a situation where reward maximization is all that matters. But if there's any epistemic uncertainty, like even uncertainty how to achieve that goal, you would want to have the epistemic value. Yeah, thank you for the answer. >> Yeah, and and a a great um ran way I'll put it in the chat. this model stream is probably in this paper is is probably the best connection between reinforcement learning and active inference. So what's so interesting is and also in relationship to to this figure it turns out that um if you are dealing with a fully observable situation then you you you can specify yeah if there's no uncertainty with what needs to be done or the situation there. If you eradicate the epistemic components of the um problem, you basically do reduce it down to just a utility maximization. But if there's uncertainty including in parameters of the agent like what learning rate is the best or what forgetting rate or just other aspects of bounded rationality within the problem like partial observability you do need an epistemic component. But then if all you have is just utility, that's where people try to make um like a learning bonus or um a ran way the same author as the as is on that stream. So yes, if one is certain about the reward, one will get the reward. Not sure how that will map to this though. Yeah. Yeah. Like in the T-mas later in seven. If the mouse knows where the food is, they don't need to do epistemic foraging. So if that's the situation that one is working in where they have, you know, an all knowing mouse, then you get the special case. Yes. Um and so that's why that's why we are um fans of the more general analytical setting that lets us describe the pure utility case or the pure epistemic case with the novelty motivation or modulation or blends of them. Alexandra. >> Okay. So, related to the the perfect mouse, um it is there a way to um either formally instruct a an agent to um or not. But but basically having an agent that goes from um taking action to increase epistemic value at the beginning of an experiment and

then changing to taking action to pragmatic value you know at the end of the experiment so that they basically get the both best of both worlds. Is there a way to formalize that in this model or is that just like imply that that is what's happening? um one example of that is this epistemic chaining which is very similar to the um T-mas but basically there are um there's different cues there's different Q locations and then there is different locations where there's going to be reward or not. So it's like the only utility function is to get the food. So then there is the consideration, okay, there's all these places I could go. I could have an uncertain I could have a a low probability of just getting the food or there are some trajectories of navigation which includes stopping by locations that I expect will reduce my uncertainty and then how much it's going to deflect out of its pragmatic path and how sensitive is it that is all the work of of the fine-tuning and the the um developing the different diagnostics for the model and those are the hard like o over in LLM world you know you'll you'll be dealing with large training data sets and different ways to set your learning rates on different examples but one piece that's easy to forget is like nowhere in here are we talking about using training data This is all just based upon the parametric operation of these graphical models. Those parameters could be developed from a training data set. That's called parametric empirical base where you basically jumpstart your bayesian inference with an empirical estimator. But you don't need to do that at all. But and then yeah like trying to find models or find the right the ranges of parameter space that would um be exploiting when it was smart to exploit and then exploring when it was smart to explore. That would be the question. Then you'd realize well we we need some we know that we need something that is going to modulate between those two. That's why we want this formalism. Okay, Benjamin, what if you just randomize the parameters? It works but more costly. Yeah, which random distribution though? That's the first question. And then the second question is you'd want to be visiting the space proportional to how good those solutions were, which is exactly the important sampling. So it's like that's the kind of like oraoris moment is like we would want to take informative actions and useful ones and if those are one and the same then then it doesn't matter. Well then we could have just you know if if we could have opened the door with our left or right hand then it doesn't matter but if we but it but then where they differ we want different sensitivities to where they differ. You could you could say my preference for ice cream is so high that that I don't explore on the path. Or you could say I have a tiny preference for this but not that. So then what's the state space? And then how would you explore that? And then you'd want to sample from that state space according to and then according to what? And then the reward the reward answer is according

to a value function. You want to spend time in rewarding locations. But that begets this whole secondary sequence of well how did you get that? And it's either well we trained it from past data or we specified it ad hoc. The active inference answer is spend time proportional to the likelihood which is tautological but that's the physics grounded. It's like the the the molecule in the energy well or the ball in the well spends time proportional to its likelihood of being in that location based upon the energy function. So that's the tautology that dispenses with the need for the reward function. But that's definitely the trick. But it puts it moves all the focus of what we actually do need to do onto the generative model which is why it's right there between the low and the high road. And that's why this is the crux of the matter is what is the structure and that's what chapter six asks about. What's the situation? What are you really trying to ask? What is the generative model? How are you going to handle time? How are you going to handle state spaces? >> Well, that's kind of why I asked my question is like in that because I peaked at that chapter. It's like well that's where the artistry comes from or that's where like the model maker needs to do some work. But I guess my question is like if you just randomize all that intuitively, it seems to me like you would get like a useless model because of its autology. If it's a tautological and you don't put any work into your like initial conditions based on like the model designer's empirical intuition or experience, then like I guess my question is like it can't it can't work, right? That's kind of my question. There has to be some I think uh bootstrapping based on either like the model creators or like empirical experience or some other sampling of empirical data. But >> I hear you also saying no, it could work out if you like randomize even the value function or something like >> that's like the kind of AGI angle that people go. They try to pull back to what's the most minimal or the generic or like what is like the generator that will generate the like all of that. But if you don't specify anything, there's not going to be a generative model. If you do specify it, it's going to be whatever it is. Those are the two situations. Al um does it make sense to look at reward in relation to precision confidence uncertainty prediction error decrease in the active inference framework? Yeah, I mean there's there's so many ways to to go with that. Like you could say you could make a model that prefers to reduce its error. Another approach is people um associate error or more error than expected with negative valence and then so then then an improving model or a model that's improving faster than expected is is associated with a effect. And then that then you can like you can have a kind of ruminating agent that looks ahead and can and then looks at like the variability or the overall um attainment of different paths and then brings that valence or uncertainty into the moments and it's like that's not necessarily computationally

what you want. It doesn't also necessarily perfectly mapped to human experience, but it's like now there's something on the table and then people can make variance and explore and say, well, in this case, but um yeah, I mean there's just so many ways to go with it. It's just like no specific linear regression how well it fits or doesn't will ever kind of cast doubt on the intrinsic coherence of the linear regression framework. So no specific generative model fitting well or not will ever make a general claim about the intrinsic coherence of the math of the generative models. And then some people think that that's like disappointed because they want to verify the analytical framework based upon a model or they want to um know that a really well-grounded mathematical basis is going to lead to like a successful startup or a drone algorithm that works. But it's like all the work is on the model creation. Active inference by itself is just these formalisms that give us a kind of controlled language to express But then which diagnostics and trade-offs and the structure of those models is is 100% on the the creator. Okay. 4.1 is basically a little preview of chapter 7. And then 4.2 is a preview of chapter 8, but it's intervened upon by this important box about blankets and message passing. preview of chapter eight with the generalized coordinates of motion Taylor series expansion. So like if if your if the real function were just a quadratic then the Taylor series expansion into the quadratic would explain 100% of the variance and further derivatives would add no more value. Whereas if it was a sinusoid then the Taylor series you would need to take out infinitely far. But if it was a sinusoid you should then pull back to recognize that you should use a sinusoidal generative model and then you'd be back to having one parameter model that would fit all the data. So that's four It rehearses some fundamentals. Bayes equation, Jensen's inequality, the properties of logs, causal modeling on a graphical setting. Then we get introduced to the Rosetta Stone 4.3 preview of chapter 7 and 8. Section 4.4 is a preview of chapter 7. Section 4.5 is a preview of chapter 8. There are some boxes with some technical topics. Markoff blankets, message passing, belief propagation, generalized coordinates, least squares approximation. So it's quite a firecracker of a as as they see it. You know, again, there's there's so many papers before and after and other documents and you can ask an LLM, etc. But they felt that this chapter outlines the basic formal ideas. You know, if we have a better or different way, we can publish it. We can write our own blogs and textbooks and everything like that. But this is really the core of how the authors see active inference and it sets up it's kind of the the center of the book. Chapter five is going to go into the neurophysiology. Chapter six kicks off the second half of the book with more of the applications. How do you make the recipe? Seven and eight discrete and continuous time.

Chapter nine, integration of data. Chapter 10, closing. So this is kind of this is the um central passages of the book and that's thank you Ahmed. Any last uh thought or question? Okay, thank you all. See you. See you next time. Bye.